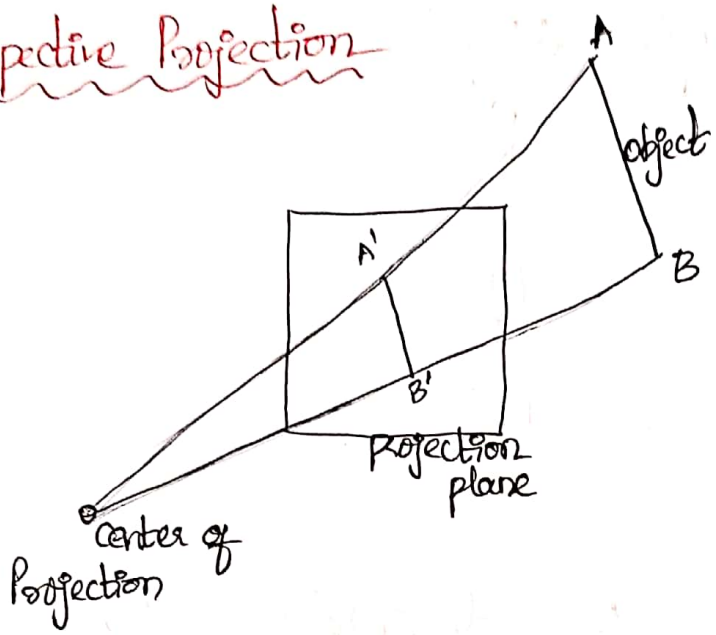
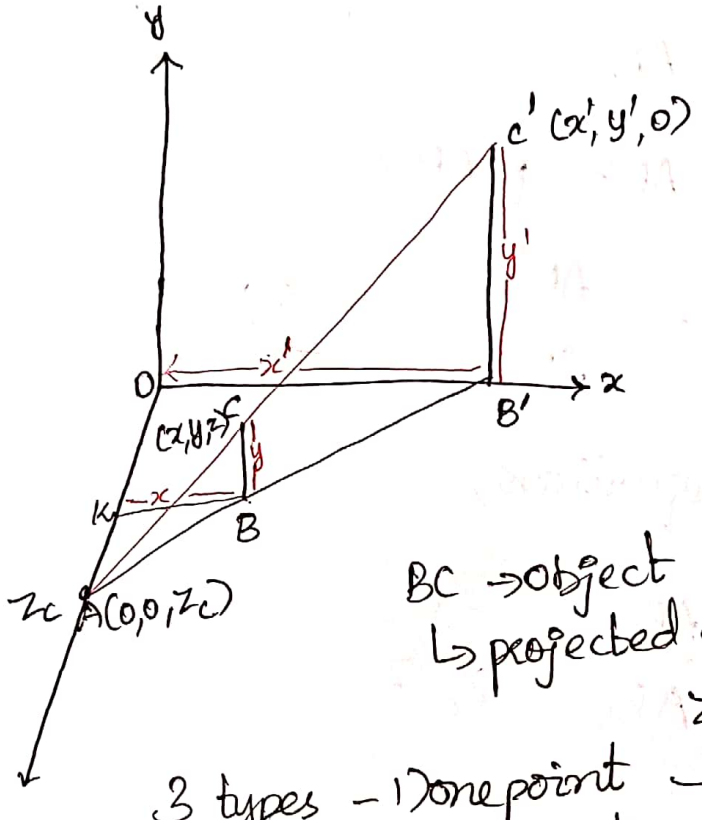
Unit - II

Perspective Projection



AB is projected on projection plane as A'B'.



BC → object
↳ projected on plane ⊥ to z axis.

3 types - 1) One point → Projection plane ⊥ to principal axis
2) Two point → Intersect 2 principal axis
3) Three point ↳ Projection plane not parallel to any principal axes.

PP is parallel to one of the principal axes

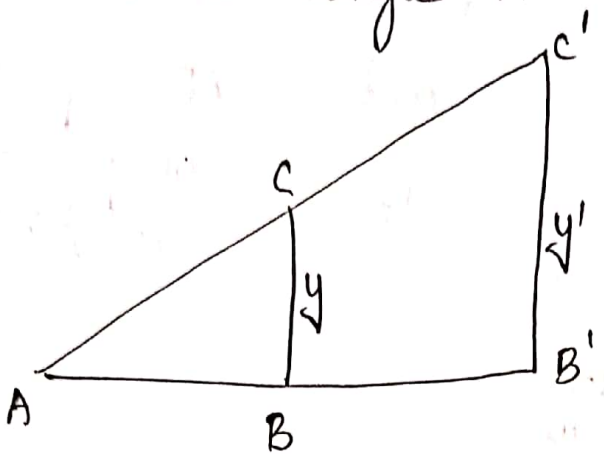
Z_c → Center of projection
↳ distance between O to A
A's coordinate is $(0, 0, Z_c)$

BC is object projected on XY plane $(x', y', 0)$.

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We have to find x' & y'

Consider the triangle $AB'C'$



$$\frac{y'}{AB'} = \frac{y}{AB} \quad \text{(triangle rule)} \rightarrow (1)$$

from ABC & $AB'C'$

$$\frac{z_c}{z_c - z} = \frac{AB'}{AB} \rightarrow (2)$$

Comparing 2 equations,

$$\frac{y'}{y} = \frac{AB'}{AB} = \frac{z_c}{z_c - z}$$

$$\frac{y'}{y} = \frac{z_c}{z_c - z}$$

$$y' = \frac{y}{1 - z/z_c}$$

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Similarly from $AB'O$

$$\frac{x'}{x} = \frac{z_c}{z_c - z}$$

$$x' = \frac{x}{1 - \frac{z}{z_c}}$$

$$\frac{x'}{AB'} = \frac{x}{AB}$$

$$\frac{x'}{x} = \frac{AB'}{AB} \rightarrow (1)$$

$$\frac{AB'}{AB} = \frac{z_c}{z_c - z} \rightarrow (2).$$

In One Point perspective projection,

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ rz+1 \end{bmatrix} \rightarrow \begin{bmatrix} x/rz+1 \\ y/rz+1 \\ z/rz+1 \\ 1 \end{bmatrix}$$

homogeneous rep.

taken in xy plane

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

projection representation

To find r,

$$x' = \frac{x}{1 - z/z_c}$$

from matrix, $x' = \frac{x}{rz+1}$

So, $\frac{x}{1 - z/z_c} = \frac{x}{rz+1}$

$$\frac{1}{\frac{z_c - z}{z_c}} = \frac{1}{rz+1}$$

$$\frac{z_c}{z_c - z} = \frac{1}{rz+1}$$

$$rzz_c + z_c = z_c - z$$

$$rzz_c = -z$$

$$r = \frac{-z}{zz_c}$$

$$r = -\frac{1}{z_c}$$

This is the value of r for the matrix in perspective projection representation

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Perspective projection

Then the matrix would be

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/z_c & 1 \end{bmatrix}$$

Center of projection $(0, 0, z_c)$

Vanishing point $(0, 0, -z_c)$

↳ Same distance on opposite side of this plane, the object will be converged at point $(0, 0, -z_c)$ called Vanishing point.

This is for $z = 0$ plane

When $x = 0$ plane $COP = (x_c, 0, 0)$

$y = 0$ plane $COP = (0, y_c, 0)$

$$p = \frac{-1}{x_c} \quad q = \frac{-1}{y_c}$$

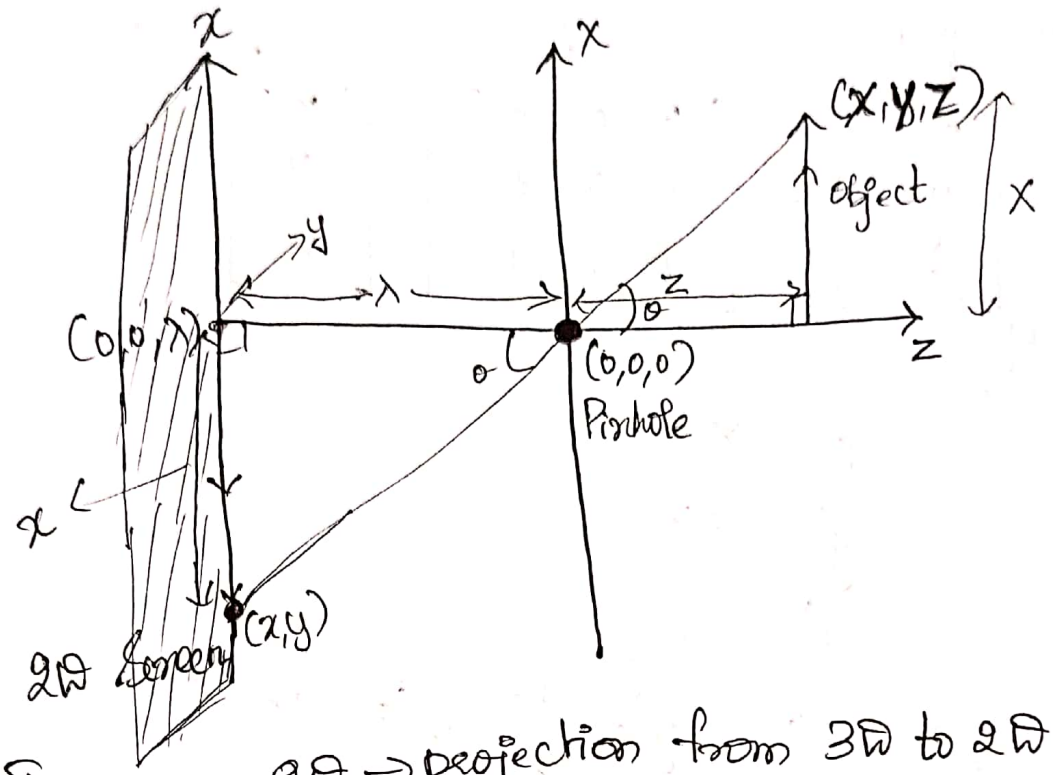
3 point Perspective Projection, the matrix would be

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & r & 1 \end{bmatrix}$$

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Camera Parameters

Camera Model



Images → 2D → projection from 3D to 2D

I/p → 3D O/p → 2D

The point (X,Y,Z) of the 3D object is projected on to the image plane of 2D (x,y)

Now we count the relationship between 3D world & 2D image plane

Consider the case of x, we have 2 triangles whose angles are equal and vertically opposite to each other same at 90°, then both triangles are similar to each other.

By similarity of triangles,

$$\frac{x}{\lambda} = \frac{X}{Z} \Rightarrow x = \frac{X\lambda}{Z}$$

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$$\frac{y}{\lambda} = \frac{Y}{Z} \Rightarrow y = \frac{Y\lambda}{Z}$$

Matrix form,

Transformation matrix transforms 3D to 2D.

3D world coordinates.

$$\begin{bmatrix} x' \\ y' \\ \omega \end{bmatrix} = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

homogenous coordinates

$$x' = X\lambda$$

$$y' = Y\lambda$$

$$\omega = Z$$

$$x' = \frac{x\lambda}{\omega} \quad y' = \frac{y\lambda}{\omega} \quad \omega = \frac{Z}{\omega}$$

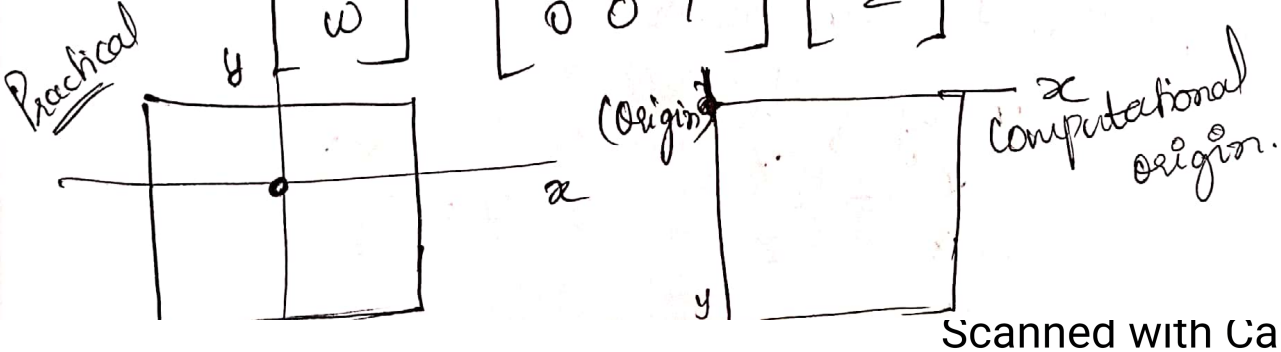
$$x' = \frac{x\lambda}{Z} \quad y' = \frac{y\lambda}{Z} \quad \omega = \frac{Z}{Z} = 1$$

Intrinsic Camera Matrix

focal length is λ so change by f.

So the matrix becomes $\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} x \\ y \\ \omega \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$



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By Simple translation

$$x = \frac{fX}{z} + \frac{P_x z}{z}$$

$$y = \frac{fy}{z} + \frac{P_y z}{z}$$

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & P_x \\ 0 & f & P_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}.$$

$$x' = fx + P_x z$$

$$x = \frac{x'}{w} = \frac{fx + P_x z}{z}$$

Cmos limitation

S → skew factor

Skew for pixels when camera used cmos. Inorder to make pixels straight ie) used for calibration, we include S factor.

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & s & P_x \\ 0 & f & P_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

3D (X, Y, Z)

(2D)
(x, y)

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & s & P_x & 0 \\ 0 & f & P_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

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Extrinsic Camera Matrix

We can extend the above thing as.

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & s & P_x \\ 0 & f & P_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Intrinsic matrix

$$x_{c_{3\times1}} = K_{3\times3} \left[I_{3\times3} \mid 0_{3\times1} \right]_{3\times4} X_w$$

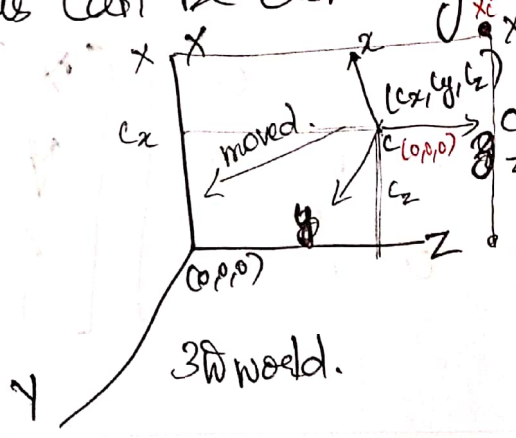
[x=0 and z=0]

The origin of 3D world is mapped on to the origin of the 2D world. (Assumption)

This may or may not be true. If we have two cameras, taking two views, which have different coordinates. So the above eqns may not work for the second camera

So we have to allign, world coordinate to the camera coordinate.

This can be done by extrinsic matrix.



Assume z dummy because for alignment

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Camera Coordinates has to be moved on to the 3D world coordinates. (Translation)

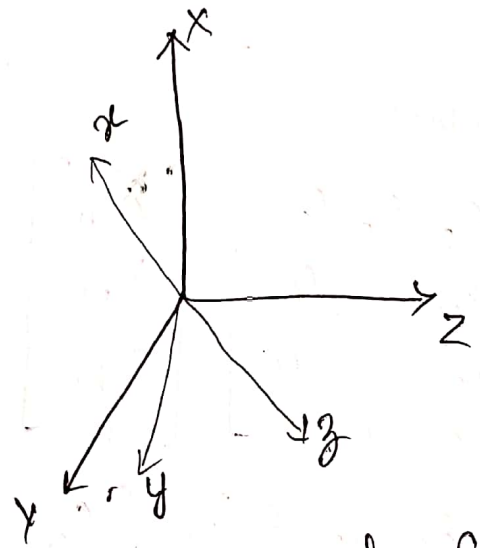
Let C → vector (C_x, C_y, C_z)

Consider any point in 3D world as X_w. which is also a vector $[X, Y, Z]$.

We have to see X_w point with respect to the camera coordinates C

$$X'_{c_{3\times1}} = X_{w_{3\times1}} - C_{3\times1}$$

So now alignment will be done.



Still alignment is required, so we require rotation matrix. In 3D, we get 3 rotation matrix

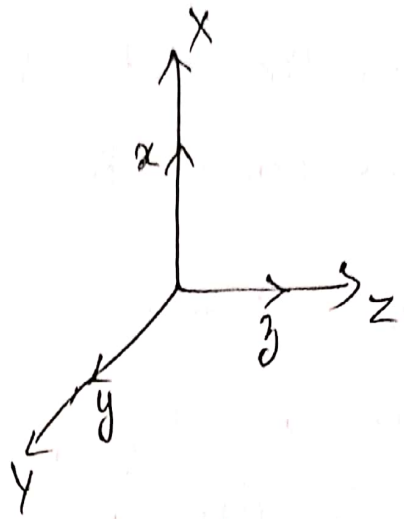
$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

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$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$



$$X_c = R \cdot X_c'$$

$$X_c = R(X_w - C)$$

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - \begin{bmatrix} C_x \\ C_y \\ C_z \end{bmatrix} \right)$$

Converting to homogeneous

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} -$$

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_x \\ C_y \\ C_z \\ 1 \end{bmatrix}$$

Compress!

$$\begin{bmatrix} X_{c\,3\times1} \\ 1 \end{bmatrix} = \begin{bmatrix} R_{3\times3} & -RC_{3\times4} \\ 0_{1\times3} & 1_{1\times1} \end{bmatrix} \begin{bmatrix} X_{w\,3\times1} \\ 1 \end{bmatrix}$$

Extrinsic Camera matrix

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$$X_{c_{3\times1}} = R_{3\times3} X_{w_{3\times1}} - R C_{3\times4}$$

$$1 = O_{1\times3} X_{w_{3\times1}} + 1_{1\times1}. \quad \text{(dummy)}.$$

Intrinsic – Transforms 3D to 2D

i/p – 3D o/p → 2D

Extrinsic – 3D to 3D. Aligns the camera coordinates and world coordinates

i/p – 3D o/p → 3D

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